

The Exact Two-Scenario Cost-Nonincrease Supremum on a Four-Terminal Planar Gadget

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Abstract

On a fixed four-terminal planar acyclic single-source unsplittable-flow gadget, we require one unsplittable routing to be no more expensive than a prescribed fractional routing under each of two positive cost scenarios. For additive upper arc deviation normalized by the largest demand, we prove that the exact non-attained supremum is $17/8$. It is not attained by any legal finite instance, but is approached by an explicit rational family whose feasible C-sets are the upward closure of $\{123,124,234\}$. A finite graph-native certificate at $\epsilon = 1/1000$ has exact value $1061/500$ and scaled unavoidable deviation 8488. The proof is self-contained; exact code verifies the finite certificate and supplies bounded regression evidence. The work is unrefereed and does not determine an unrestricted planar or many-scenario constant.

This is a public unrefereed research disclosure, first released in v0.2.0. Revision 2 corrects provenance and status wording only; it changes no theorem statement, proof conclusion, constant, certificate, or verifier result. The two hostile-review rounds were role-separated AI-assisted model critiques, not external human mathematical review. The second proof-only critic found no theorem-level gap and recommended acceptance after local revisions, which were integrated before the human release decision. No external human mathematical review is documented.

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1 Introduction

Single-source unsplittable-flow rounding converts a fractional flow into one source-terminal path per demand. The classical cost-constrained question asks for a routing that does not exceed the fractional cost while controlling arc overload. The present note keeps one compact four-terminal planar DAG fixed and imposes two simultaneous linear cost budgets. It asks for the exact worst-case normalized additive upper arc deviation on that gadget.

The model is intentionally narrow. A routing is feasible when it satisfies scenario-wise cost non-increase in both scenarios; equality is not required. The theorem does not determine the unrestricted planar constant, the otherwise-identical arbitrary one-scenario fixed-graph constant, or a many-scenario constant.

Dmitry Rybin’s public counterexample of 22 July 2026 was the direct catalyst for this research line. The earlier release in this repository established a one-scenario planar lower bound and restricted equal-cost sharpness result. The new theorem is a separate fixed-gadget two-scenario result.

2 The graph, paths, and flow model

Let the directed graph have source s , trunk vertices $v_1, \dots, v_5$, terminals $t_1, \dots, t_4$, and trunk arcs

$$a_1 = sv_1, \quad a_2 = v_1v_2, \quad a_3 = v_2v_3, \quad a_4 = v_3v_4, \quad a_5 = v_4v_5.$$

The eight terminal arcs are

$$e_1 = st_1, \quad c_1 = v_3t_1, \quad e_2 = st_2, \quad c_2 = v_5t_2, \quad e_3 = v_1t_3, \quad c_3 = v_5t_3, \quad e_4 = v_2t_4, \quad c_4 = v_4t_4.$$

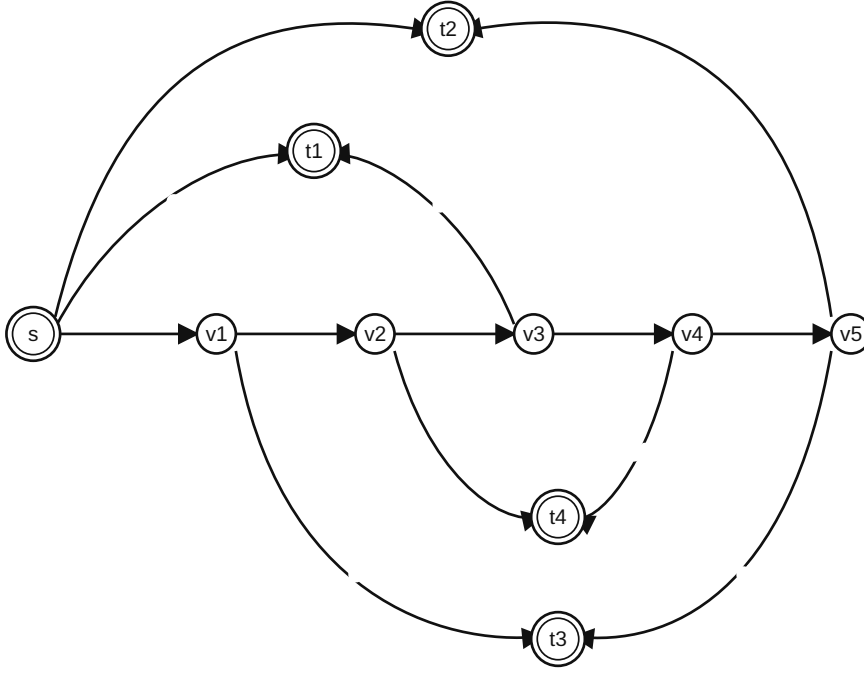


Figure 1: Released four-terminal gadget

Each terminal has exactly the following two designated source-terminal paths:

terminal i	path E_i	path C_i
1	e_1	$a_1 \ a_2 \ a_3 \ c_1$
2	e_2	$a_1 \ a_2 \ a_3 \ a_4 \ a_5 \ c_2$
3	$a_1 \ e_3$	$a_1 \ a_2 \ a_3 \ a_4 \ a_5 \ c_3$
4	$a_1 \ a_2 \ e_4$	$a_1 \ a_2 \ a_3 \ a_4 \ c_4$

Terminal i has demand $d_i > 0$. A fractional flow sends proportion $p_i \in [0, 1]$ on C_i and

$$q_i := 1 - p_i$$

on E_i . With $\mathbf{1}_P(a)$ denoting arc membership in path P , its arc load is

$$x(a) = \sum_{i=1}^4 d_i (p_i \mathbf{1}_{C_i}(a) + q_i \mathbf{1}_{E_i}(a)). \quad (1)$$

An unsplittable routing is described by its E -set $R \subseteq [4]$: terminals in R use E , and the others use C . Its load is

$$y^R(a) = \sum_{i \notin R} d_i \mathbf{1}_{C_i}(a) + \sum_{i \in R} d_i \mathbf{1}_{E_i}(a). \quad (2)$$

3 Scenario-wise cost non-increase

For scenario $j \in \{1, 2\}$, let $c_a^{(j)} \geq 0$ be the per-unit arc cost. Write

$$\gamma_{i,C}^{(j)} := d_i \sum_{a \in C_i} c_a^{(j)}, \quad \gamma_{i,E}^{(j)} := d_i \sum_{a \in E_i} c_a^{(j)},$$

and assume the full-demand E -minus- C differences are strictly positive:

$$k_i^{(j)} := \gamma_{i,E}^{(j)} - \gamma_{i,C}^{(j)} > 0. \quad (3)$$

Conversely, every positive difference vector is realizable on this graph: put per-unit cost $k_i^{(j)}/d_i$ on the private E -only arc e_i and zero on every other arc. Thus optimizing over positive difference vectors is equivalent to optimizing over legal nonnegative scenario arc costs of this form.

Let $C_j(z) = \sum_a c_a^{(j)} z(a)$. Direct expansion gives

$$\begin{aligned} C_j(y^R) - C_j(x) &= \sum_{i \in R} k_i^{(j)} - \sum_i q_i k_i^{(j)} \\ &= k^{(j)}(R) - k^{(j)} \cdot q. \end{aligned} \quad (4)$$

Accordingly, a routing is **scenario-wise cost non-increasing** precisely when

$$C_j(y^R) \leq C_j(x) \iff k^{(j)}(R) \leq k^{(j)} \cdot q \quad (j = 1, 2). \quad (5)$$

Equation (5) is an inequality. It does **not** require equality between the fractional and unsplittable scenario costs. In this paper, “respects both scenario budgets” always means (5).

Put

$$d_{\max} := \max_i d_i.$$

The two-scenario fixed-topology constant is

$$\beta_G^{(2\text{sc})} := \sup \frac{1}{d_{\max}} \min_{R \text{ satisfying (5)}} \max_{a \in \mathcal{A}(G)} (y^R(a) - x(a)), \quad (6)$$

where the supremum ranges over positive demands, fractional proportions, and two positive difference vectors. This is a normalized **additive upper arc-deviation** constant, not a multiplicative total-cost or capacity ratio.

4 Main theorem

4.1 Theorem RB-003

$$\beta_G^{(2\text{sc})} = \frac{17}{8} = 2.125.$$

The same supremum is obtained over rational data. No legal finite instance attains $17/8$; rational instances with intrinsic feasible C-family

$$\uparrow \{123, 124, 234\} \quad (7)$$

approach it.

The family in (7) is called **F126** in the earlier one-scenario census, but the identifier is not needed for this theorem: its feasible C-sets are exactly

$$123, \quad 124, \quad 234, \quad 1234.$$

5 E-side normalization and graph identities

If $q = 0$, the fractional flow is already all-C and has zero deviation. Otherwise define

$$B_j := k^{(j)} \cdot q > 0, \quad w_i^{(j)} := \frac{k_i^{(j)}}{B_j}.$$

Then

$$\sum_i q_i w_i^{(j)} = 1, \quad (8)$$

and an E-set R respects both budgets exactly when

$$w^{(1)}(R) \leq 1, \quad w^{(2)}(R) \leq 1. \quad (9)$$

Normalize $d_{\max} = 1$ and put

$$h_i := d_i q_i, \quad H := \sum_i h_i. \quad (10)$$

The trunk path-difference supports $I_i = C_i \setminus E_i$ are

$$\begin{aligned} I_1 &= \{a_1, a_2, a_3\}, & I_2 &= \{a_1, a_2, a_3, a_4, a_5\}, \\ I_3 &= \{a_2, a_3, a_4, a_5\}, & I_4 &= \{a_3, a_4\}. \end{aligned} \tag{11}$$

Equivalently, the terminals contributing positively to the five all-C trunk deviations are

$$\begin{aligned} J_1 &= \{1, 2\}, & J_2 &= \{1, 2, 3\}, & J_3 &= \{1, 2, 3, 4\}, \\ J_4 &= \{2, 3, 4\}, & J_5 &= \{2, 3\}. \end{aligned} \tag{12}$$

Every terminal-private arc is used by only one route of one terminal, so every positive private-arc deviation is at most $d_i \leq 1$.

The all-C routing has trunk maximum H . If singleton E-set $\{r\}$ is feasible, direct substitution in (12) gives:

r	maximum trunk deviation
1	$H - h_1$
2	$H - d_2$
3	$\max\{h_1 + h_2, H - d_3\}$
4	$H - h_4$.

(13)

If a feasible E-pair exists, its complementary C-set has size two. Routing that C-pair creates positive trunk contribution at most two and positive private contribution at most one. Hence only instances with **no feasible E-pair** can have value above two.

6 Two auxiliary lemmas

Call a singleton or pair T **blocked by scenario j** when

$$w^{(j)}(T) > 1.$$

When no E-pair is feasible, every pair is blocked by at least one scenario. Every singleton not feasible under both scenarios is also blocked by at least one scenario. Assign each such blocked set to one scenario that blocks it.

6.1 Lemma 1 — two-colour fractional-matching bound

For one scenario j , let $\lambda_T \geq 0$ be weights on its assigned blocked singletons and pairs, with vertex loads

$$\sum_{T \ni i} \lambda_T \leq h_i.$$

Then

$$\sum_T \lambda_T < 1.$$

Proof. If all λ_T vanish, the conclusion is immediate. Otherwise every positively weighted assigned set has $w^{(j)}(T) > 1$, so

$$\sum_T \lambda_T < \sum_T \lambda_T w^{(j)}(T) = \sum_i w_i^{(j)} \sum_{T \ni i} \lambda_T \leq \sum_i w_i^{(j)} h_i \leq \sum_i w_i^{(j)} q_i = 1.$$

The penultimate inequality uses $h_i = d_i q_i \leq q_i$. $\square$

Assign every blocked singleton and pair to one scenario that blocks it. If λ_T is any weighting of the assigned sets satisfying

$$\sum_{T \ni i} \lambda_T \leq h_i$$

for every vertex, then its restriction to either scenario satisfies the same vertex-load bounds. Lemma 1 gives total weight below one in each colour, so

$$\sum_T \lambda_T < 2.$$

6.2 Lemma 2 — pair-blocked three-set, with arbitrary unit-bounded profits

Let T be a three-element terminal set. Suppose one scenario blocks every pair in T . Then, for every profit vector $c_i \in [0, 1]$,

$$\sum_{i \in T} c_i q_i \leq 2. \tag{14}$$

Proof. The restricted vector $(q_i)_{i \in T}$ is feasible for

$$\max \left\{ \sum_{i \in T} c_i z_i : \sum_{i \in T} w_i^{(j)} z_i \leq 1, \ 0 \leq z_i \leq 1 \right\},$$

because all omitted terms in (8) are nonnegative. A box-constrained one-knapsack LP has an extreme optimum with at most one fractional coordinate: if the knapsack inequality is slack, all coordinates of an extreme point are at box bounds; if it is tight, at least two independent box constraints must also be active. Its integral part contains at most one selected item, since every pair is blocked. That integral item contributes at most one, and the possible fractional item contributes at most one. Therefore the optimum, and hence the value at $\mathbf{q}$, is at most two. $\square$

Two consequences will be used separately:

$$\sum_{i \in T} d_i q_i \leq 2 \quad \text{by taking } c_i = d_i, \quad (15)$$

and

$$\sum_{i \in T} q_i \leq 2 \quad \text{by taking } c_i = 1. \quad (16)$$

If a feasible singleton E-set $\{r\}$ has complement T , and one scenario blocks every pair in T , then (15) shows that routing $\mathbf{r}$ on E has positive trunk deviation at most two; private deviations remain at most one.

7 Upper bound: reduction to a star-triangle

Let t be the minimum maximum deviation among budget-respecting routings. Assume $t > 2$, since otherwise the desired upper bound is immediate. The all- C route is feasible, so $H \geq t$. Put

$$\Delta := H - t \geq 0. \quad (17)$$

Let

$$A := \{i : \{i\} \text{ is feasible under both scenarios}\}$$

be the individually omittable terminals. From (13), every i in A satisfies

$$h_i \leq \Delta \quad (i = 1, 4), \quad d_i \leq \Delta \quad (i = 2, 3). \quad (18)$$

For terminal 3, this uses $t > 2$ and $h_1 + h_2 \leq 2$: the first term in its row of (13) cannot reach $\mathfrak{t}$, so $H - d_3 \geq t$. In particular, $h_i \leq \Delta$ for every i in A .

7.1 At least three individually omittable terminals

If $|A| = 0$, put matching weight h_i on each blocked singleton. Its total is $H > 2$, contradicting Lemma 1 across the two colours.

If $|A| = 1$, put weight h_i on the three blocked singletons. The total is at least

$$H - \Delta = t > 2,$$

again a contradiction.

If $A = \{r, s\}$, put singleton weights on the other two vertices and weight $\min(h_r, h_s)$ on the blocked pair $\{r, s\}$. The total is

$$H - h_r - h_s + \min(h_r, h_s) = H - \max(h_r, h_s) \geq H - \Delta = t > 2,$$

again contradicting Lemma 1. Hence $|A| \geq 3$.

7.2 All four terminals individually omittable

For one scenario, form the graph whose edges are its blocked pairs. If this graph is triangle-free, it is a subgraph of a star: choose a maximum-weight vertex v ; if a blocked edge ij avoided v , then vi and vj would also be blocked, creating a triangle.

Two stars cannot cover all six edges of K_4 . Since every pair is blocked by at least one scenario, one scenario contains a blocked triangle. Its complementary terminal lies in A , and (15) gives a route of value at most two. Thus $|A| = 4$ is not hard.

7.3 Exactly three terminals individually omittable

Let u be the unique terminal outside A . If one scenario has a blocked triangle whose complementary terminal lies in A , (15) again gives a route of value at most two.

Otherwise choose a scenario that blocks singleton $\{u\}$. Positivity makes it block every pair incident with u . It cannot block any pair internal to A , because that pair together with u would form a blocked triangle whose complement lies in A . Therefore the other scenario must block all three pairs internal to A .

Up to exchanging scenarios, the forced blocker core is:

- one scenario blocks u and all three pairs incident with it, and blocks no pair internal to A — a **star**;
- the other blocks every pair inside A — a **triangle**.

The triangle scenario may also block one additional pair incident with u ; this does not affect the argument. Two such additional incident pairs would create a blocked triangle whose complementary terminal lies in A , the case already excluded above.

By (16) and $h_u \leq 1$,

$$\sum_{i \in A} q_i \leq 2, \quad h_u \leq 1. \quad (19)$$

8 Exact star-triangle optimization

8.1 The non-omittable terminal is central: $u \in \{2, 3\}$

Then A contains the two outer terminals a, b in $\{1, 4\}$ and one remaining central terminal c . For $0 \leq \Delta \leq 1$, (18) and (19) give

$$H \leq 1 + \min\{q_a, \Delta\} + \min\{q_b, \Delta\} + \Delta q_c, \quad q_a + q_b + q_c \leq 2. \quad (20)$$

The exact envelope is

$$H \leq \begin{cases} 1 + 3\Delta, & 0 \leq \Delta \leq \frac{1}{2}, \\ 1 + 4\Delta - 2\Delta^2, & \frac{1}{2} \leq \Delta \leq 1. \end{cases} \quad (21)$$

Indeed, the two outer coordinates have marginal value one until each reaches Δ , while the central coordinate has marginal value Δ . Consequently,

$$t = H - \Delta \leq \begin{cases} 1 + 2\Delta \leq 2, & 0 \leq \Delta \leq \frac{1}{2}, \\ 1 + 3\Delta - 2\Delta^2 = \frac{17}{8} - 2\left(\Delta - \frac{3}{4}\right)^2 \leq \frac{17}{8}, & \frac{1}{2} \leq \Delta \leq 1. \end{cases} \quad (22)$$

If $\Delta \geq 1$, then (19) gives $H \leq 3$, hence $t = H - \Delta \leq 2$.

8.2 The non-omittable terminal is outer: $u \in \{1, 4\}$

Now $\mathbf{A}$ contains one outer terminal $\mathbf{a}$ and two central terminals $\mathbf{b}, \mathbf{c}$. For $0 \leq \Delta \leq 1$,

$$H \leq 1 + \min\{q_a, \Delta\} + \Delta(q_b + q_c), \quad q_a + q_b + q_c \leq 2.$$

Allocating the budget first to the outer coordinate gives

$$H \leq 1 + 3\Delta - \Delta^2,$$

so

$$t \leq 1 + 2\Delta - \Delta^2 = 2 - (1 - \Delta)^2 \leq 2. \quad (23)$$

Again $\Delta \geq 1$ gives $t \leq 2$. Equations (22) and (23) prove

$$\beta_G^{(2\text{sc})} \leq \frac{17}{8}. \quad (24)$$

9 Non-attainment

9.1 Corollary RB-003a

Every legal finite instance has value strictly below $17/8$.

Proof. Suppose a finite instance attained $t = 17/8$. Since this exceeds two, the proof above forces the central star-triangle branch. Equality in (22) forces

$$\Delta = \frac{3}{4}, \quad H = t + \Delta = \frac{23}{8}.$$

At $\Delta = 3/4$, let $\mathbf{a}, \mathbf{b}$ be the two outer terminals in $\mathbf{A}$ and let $\mathbf{c}$ be the remaining central terminal. Equality throughout (20)–(22) gives

$$\frac{23}{8} = H = h_u + h_a + h_b + h_c \leq 1 + \min\{q_a, \frac{3}{4}\} + \min\{q_b, \frac{3}{4}\} + \frac{3}{4}q_c \leq \frac{23}{8}.$$

Hence every inequality in this chain is tight, in particular $h_u = 1$. Because $h_u = d_u q_u$ and both factors are at most one,

$$d_u = q_u = 1.$$

But u is not individually omittable, so some scenario has $w_u^{(j)} > 1$. Normalization (8) then yields

$$1 = \sum_i q_i w_i^{(j)} \geq q_u w_u^{(j)} = w_u^{(j)} > 1,$$

which is impossible. $\square$

Thus F126 supplies an extremizing sequence, not a finite extremizer.

10 Rational extremizing sequence

Let $0 < \varepsilon < 1/4$ be rational, take the non-omittable central terminal $u = 2$, and set

$$\begin{aligned} q &= \left(\frac{3}{4} - \varepsilon, 1 - \varepsilon, \frac{1}{2}, \frac{3}{4} - \varepsilon \right), \\ p &= 1 - q = \left(\frac{1}{4} + \varepsilon, \varepsilon, \frac{1}{2}, \frac{1}{4} + \varepsilon \right), \\ d &= \left(1, 1, \frac{3}{4}, 1 \right). \end{aligned} \tag{25}$$

Use positive difference vectors

$$k^{(1)} = \left(1, \frac{3}{\varepsilon}, 1, 1 \right), \quad k^{(2)} = (1, \varepsilon, 1, 1). \tag{26}$$

Their fractional budgets are

$$B_1 = \frac{3}{\varepsilon} - 1 - 2\varepsilon, \quad B_2 = 2 - \varepsilon - \varepsilon^2.$$

Scenario 1 blocks terminal 2 by itself and permits every subset of $A = \{1, 3, 4\}$. Scenario 2 permits each singleton in $\mathbf{A}$ but blocks every pair in $\mathbf{A}$. Therefore the E-sets respecting both budgets are exactly

$$\emptyset, \quad \{1\}, \quad \{3\}, \quad \{4\},$$

and the feasible C-family is intrinsically the family (7).

Here

$$h = \left(\frac{3}{4} - \varepsilon, 1 - \varepsilon, \frac{3}{8}, \frac{3}{4} - \varepsilon \right), \quad H = \frac{23}{8} - 3\varepsilon.$$

Exact evaluation of the four feasible routings gives

E-set	maximum deviation	
$\emptyset$	$\frac{23}{8} - 3\varepsilon$	(27)
$\{1\}$	$\frac{17}{8} - 2\varepsilon$	
$\{3\}$	$\frac{17}{8} - 3\varepsilon$	
$\{4\}$	$\frac{17}{8} - 2\varepsilon$	

Thus the exact finite objective is

$$\frac{17}{8} - 3\varepsilon,$$

which approaches $17/8$ through rational instances. Together with (24), this proves Theorem RB-003.

11 Concrete finite certificate

At $\varepsilon = 1/1000$,

$$p = \left(\frac{251}{1000}, \frac{1}{1000}, \frac{1}{2}, \frac{251}{1000} \right), \quad d = (1, 1, 3/4, 1),$$

with integer differences

$$k^{(1)} = (1, 3000, 1, 1), \quad k^{(2)} = (1000, 1, 1000, 1000).$$

The fractional budgets are

$$B_1 = \frac{1499499}{500} = 2998.998, \quad B_2 = \frac{1998999}{1000} = 1998.999.$$

Every feasible unsplittable routing has scenario costs (0 or 1) in scenario 1 and (0 or 1000) in scenario 2. These are below the fractional budgets; they are not equal to them. This is exactly the non-increase semantics in (5).

The exact normalized objective is

$$\frac{1061}{500} = 2.122.$$

After multiplying demands by 4000, the maximum demand is 4000 and every route respecting both budgets has upper deviation at least 8488. The graph-native certificate enumerates all 16 routings and all 13 arcs.

Legal nonnegative arc costs realize each scenario by placing per-unit cost $k_i^{(j)}/d_i$ on terminal i 's private E-only arc and zero elsewhere.

12 Computational evidence and its limits

The human-readable proof above is authoritative. The package’s computations serve four narrow roles:

1. **Exact finite certificate:** enumerate the 16 routings and 13 arcs of the concrete lower instance and verify (4)-(5) exactly.
2. **Exact four-label threshold recognition:** enumerate all 168 downsets, provide positive integer threshold witnesses for 149, exact two-trade impossibility certificates for the other 18 nonempty downsets, and identify the empty inadmissible family.
3. **Abstract blocker regression:** classify the 11,175 unordered pairs of the 149 scalar threshold patterns. These abstract pairs need not share one baseline $\mathbf{q}$; this census is not a shared-baseline realization theorem and is not used to prove the analytic upper bound.
4. **Envelope grid regression:** evaluate the already-proved envelopes on a denominator-16 grid. This is not continuous optimization and is not a proof of (21)-(23).

The secondary script is a separate code path, not an independent mathematical derivation: it shares the same graph support matrix, blocker structure, and lower-family ansatz.

13 Context, scope, and nonclaims

The previously released restricted one-control benchmark is

$$L = \frac{299 - 41\sqrt{41}}{32} = 1.139747070789 \dots$$

Arithmetic gives $17/8 > L$, but this is **not** a controlled marginal estimate of the effect of adding a second scenario. The released value L belongs to a restricted equal-full-cost, all-pairs-feasible one-scenario regime, whereas RB-003 ranges over two arbitrary positive difference vectors. The exact otherwise-identical arbitrary one-scenario constant is not established here. See `BASELINE_CONTEXT_AND_DEPENDENCIES.md`.

The theorem also does not establish that a particular sequential algorithm is suboptimal. It shows that the intersection of two non-increase constraints can support a star-triangle obstruction with supremum $17/8$; algorithmic comparisons require separate analysis.

The finite certificate uses within-scenario weight ratios 3000 and 1000, and the extremizing sequence requires unbounded ratios as $\varepsilon \downarrow 0$. The bounded-heterogeneity constant

$$\beta_G^{(2\text{sc})}(\kappa), \quad \frac{\max_i k_i^{(j)}}{\min_i k_i^{(j)}} \leq \kappa,$$

is a separate, commercially important problem.

RB-003 is restricted to this fixed graph, two positive scenario-wise cost-nonincrease constraints, and the additive upper-deviation objective. It does not determine an unrestricted planar constant, a many-scenario constant, a bounded-condition-number constant, or a multiplicative capacity-augmentation factor.

14 Related work and novelty posture

The foundational single-source unsplittable-flow framework is due to Dinitz, Garg, and Goemans. Traub, Vargas Koch, and Zenklusen prove a planar cost-preserving upper bound with one cost vector, while Majthoub Almoghrabi, Skutella, and Warode obtain strong series-parallel integrality and unsplittable-flow results. Morell and Skutella study simultaneous arc-wise lower and upper load bounds. Multicriteria, robust, and reroutable flow models supply important adjacent context, but use different constraints or uncertainty semantics.

The four-label threshold-family calculations used in the finite registry overlap classical weighted-simple-game and threshold-function theory; no novelty claim is made for that general classification. A targeted search on 28 July 2026 found no indexed match for the exact fixed-gadget, shared-baseline two-scenario formulation, the star-triangle proof, or the value $17/8$. This is a limited search result, not exhaustive novelty clearance.

15 Research provenance, validation, and stewardship

Matthew Protti selected and directed the research program, set the verification and claim standards, repeatedly challenged intermediate conclusions, approved the final scope, and accepts responsibility for the release. OpenAI GPT-5.6 Pro generated and developed substantial portions of the construction, proof, verification code, adversarial analysis, and manuscript. The documented hostile-review rounds were role-separated AI-assisted model critiques. Separate model sessions and Codex implementations reduce some shared-error risk but are not independent human mathematical review or peer review.

The proof above is authoritative. The finite programs establish or corroborate only their declared finite propositions. Agent self-assessment is not acceptance evidence. The repository includes an intervention record, checker-assurance statement, deterministic replay, and correction policy.

OpenAI’s 28 July 2026 exploratory field report on agentic scientific computing describes a broad shift from implementation toward human specification, validation, orchestration, and stewardship. It is cited here as contemporaneous methodological context, not as validation or endorsement of this result.

16 Acknowledgements

Dmitry Rybin’s 22 July 2026 counterexample was the direct starting point for this research line. Role-separated AI-assisted critique records exposed scope, proof, and presentation defects before the human release decision.

17 References

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